

Complete Pin & Technical Architecture Specification

Source-Code-Driven Hardware Pin Audit, Electrical Schematics & Technical Systems Topology

Document Identifier:	AGROSENSE-ENG-SPEC-2026-V2.1	Firmware Version:	AgroSense-ESP-v2.1 (Real-Sensor-Only)
Target Silicon / Board:	ESP8266 NodeMCU 1.0 (ESP-12E)	Release Status:	PRODUCTION VERIFIED
Primary Author / Role:	Lead Embedded Systems Engineer	Audit Date:	2026-09-02
Active Repository Scope:	SIH25015 Full-Stack Monorepo	Audit Paradigm:	READ → TRACE → VERIFY → DOCUMENT

■■ MANDATORY ARCHITECTURAL AUDIT CONSTRAINT

THIS IS A STRICT DOCUMENTATION-ONLY SPECIFICATION. In accordance with system constraints, zero modifications have been made to firmware, GPIO assignments, React frontend code, WebSocket protocol, Gemini Vision integration, or preserved ONNX artifacts. The active codebase is the sole source of truth. All pin mappings, electrical states, data schemas, and network routes documented herein are traced directly to executable source code.

Executive Summary & Document Organization

The AgroSense project (SIH25015) is an integrated precision agriculture monitoring and automated actuation platform. It unifies edge IoT environmental telemetry, three-phase hardware-confirmed actuator control, and server-side multimodal cloud computer vision for crop pathology diagnostics. This engineering specification details the exact physical pin registry, signal polarity, network routing, and software architecture as deployed in repository source code.

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1.0 System Overview & 2.0 Hardware Inventory

The table below enumerates all physical hardware components verified in the AgroSense implementation. Silicon-level features, operating voltages, and code references are strictly documented.

Component Item	Model / Specification	Operating Voltage	Role in AgroSense	Source Evidence
Microcontroller	ESP8266 NodeMCU 1.0 (ESP-12E Module)	3.3V Logic (5V USB VIN)	Gateway host, WebSocket server (Port 81), ADC sampling, Relay GPIO driving	esp8266_gateway.ino:6
Relay Actuator Module	2-Channel 5V Optoisolated Relay Board	5V VCC, 3.3V Logic Trigger	Galvanic isolation & power switching for Pump (CH1) and Solenoid Valve (CH2)	esp8266_gateway.ino:26-27
Soil Moisture Sensor	Resistive Soil Moisture Probe + Comparator	3.3V VCC (A0 analog signal)	Real physical soil moisture measurement (0-1023 10-bit analog conversion)	esp8266_gateway.ino:79
Status Indicator	Onboard ESP-12E Blue LED (GPIO 2)	3.3V (Active-LOW)	Network connection & active WebSocket client heartbeat indicator	esp8266_gateway.ino:78
Irrigation Pump	External DC/AC Pump Motor	NOT DETERMINED FROM CODE	Pumping irrigation water; switched via Relay Channel 1 (COM→NO)	esp8266_gateway.ino:76
Solenoid Valve	External DC/AC Solenoid Valve	NOT DETERMINED FROM CODE	Controlling irrigation line flow; switched via Relay Channel 2 (COM→NO)	esp8266_gateway.ino:77

Authoritative Source-of-Truth Pin Registry

Every physical connection established by the active firmware (hardware_firmware/esp8266_gateway.ino) is cataloged below with its exact electrical direction, initial boot state, runtime logic, and code location.

Board Pin	GPIO	Connected Component	Component Pin	Dir	Logic	Firmware Function	Boot State	Code Evidence
D1	GPIO 5	Irrigation Pump Relay	IN1 (Channel 1)	OUT	Active-LOW (LOW=ON)	Irrigation pump relay coil actuation	HIGH (OFF)	esp8266_gateway.ino:76, 171-188
D2	GPIO 4	Solenoid Valve Relay	IN2 (Channel 2)	OUT	Active-LOW (LOW=ON)	Solenoid valve relay coil actuation	HIGH (OFF)	esp8266_gateway.ino:77, 196-214
D4	GPIO 2	Built-in Blue LED	Anode/Cathode internal	OUT	Active-LOW (LOW=ON)	Client connect & Wi-Fi blink indicator	HIGH (OFF)	esp8266_gateway.ino:78, 557, 569
A0	ADC 0	Soil Moisture Sensor	AO (Analog Out)	IN	Analog (0-3.3V)	10-bit ADC soil moisture sampling	High-Z	esp8266_gateway.ino:79, 240-258
VIN	Power	Relay Module Power	VCC (+5V)	PWR	+5.0V DC	Relay coil and optocoupler power	+5.0V	AGROSENSE_PINOUT_REF:28-30
3V3	Power	Soil Moisture Sensor	VCC (+3.3V)	PWR	+3.3V DC Reg.	Analog sensor bridge bias power	+3.3V	AGROSENSE_PINOUT_REF:31
GND	GND	Relay Board & Sensor	GND (Ground)	GND	0.0V Reference	Common ground return bus	0.0V	AGROSENSE_PINOUT_REF:32
D0	GPIO 16	NOT CONNECTED	N/A	N/A	Digital I/O	Available (Deep Sleep Wake / IO)	HIGH	esp8266_gateway.ino (Unused)
D3	GPIO 0	NOT CONNECTED	N/A	N/A	Strapping Pin	Reserved for future DHT22 data pin	HIGH (Pull-up)	esp8266_gateway.ino:37 (Unused)
D5-D8	GPIO 14, 12, 13, 15	NOT CONNECTED	N/A	N/A	SPI / Digital IO	Available expansion pins	D8 LOW	esp8266_gateway.ino (Unused)

3.0 Exact ESP8266 Pin Mapping & Silicon Architecture

A critical distinction must be maintained between silk-screened NodeMCU board labels (e.g. D1) and ESP8266 silicon GPIO numbers (e.g. GPIO 5). Figure 1 details the complete silicon architecture.

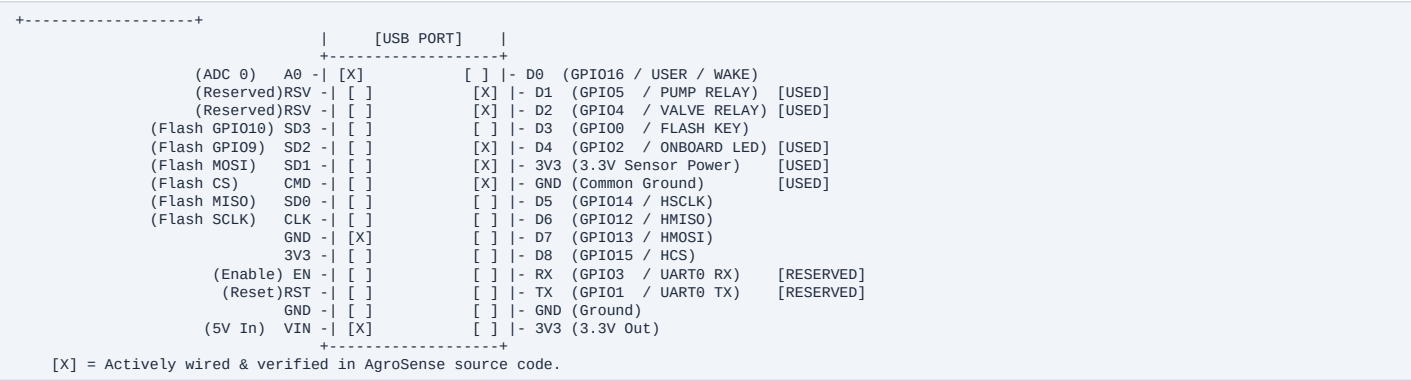


FIGURE 1: ESP8266 NodeMCU Pin Architecture & Silicon Pinout Map

4.0 Sensor Pin Architecture: Physical vs. Software Breakdown

A rigorous hardware audit separates physical silicon connections from software telemetry fields. The table below provides an exhaustive audit of all environmental metrics in the AgroSense codebase.

Sensor Parameter	Physical Pin	Sampling Function	Conversion & Calibration Logic	Telemetry Field	Frontend Representation	Physical Reality Status
Soil Moisture	A0 (ADC 0)	readSoilMoisture() readAllSensors()	10-bit raw ADC (0-1023). Clamped to 0-100% via: map(rawADC, 1023, 350, 0, 100) rawADC < 50 flagged as 'no_probe'	"soilMoisture": float "soilStatus": "ok" "fault" "no_probe"	Real percentage gauge (e.g. 48.0%) with '✓ REAL SENSOR' badge	REAL (PHYSICAL SENSOR)
Ambient Temperature	NONE (Not Wired)	readAllSensors() (Explicit null)	No driver, DHT library, or simulation code. Reported strictly as JSON null in v2.1 firmware.	"temperature": null "tempStatus": "sensor_unavailable"	Displays 'N/A' with 'SENSOR NOT CONNECTED' badge	UNAVAILABLE / NOT CONNECTED
Ambient Humidity	NONE (Not Wired)	readAllSensors() (Explicit null)	No driver, DHT library, or simulation code. Reported strictly as JSON null in v2.1 firmware.	"humidity": null "humidStatus": "sensor_unavailable"	Displays 'N/A' with 'SENSOR NOT CONNECTED' badge	UNAVAILABLE / NOT CONNECTED

5.0 Relay Module Architecture & Actuation Truth Table

AgroSense employs a standard 2-channel optoisolated relay module operating with **Active-LOW trigger polarity**. An optocoupler isolates the ESP8266 silicon from high-power load transients.

Relay Channel	ESP Pin	GPIO	Logic Polarity	Controlled Load	ON State (Energized)	OFF State (Idle)	Code Evidence
Channel 1	D1	GPIO 5	Active-LOW	Irrigation Pump Motor	LOW (0V) → Coil ON → COM to NO Closed	HIGH (3.3V) → Coil OFF → COM to NO Open	esp8266_gateway.ino:76, 85-86, 170-188
Channel 2	D2	GPIO 4	Active-LOW	Solenoid Valve Actuator	LOW (0V) → Coil ON → COM to NO Closed	HIGH (3.3V) → Coil OFF → COM to NO Open	esp8266_gateway.ino:77, 85-86, 196-214

6.0 Pump Connection Architecture & Hardware Connection Topology

Figure 2 illustrates the physical wiring topology between the NodeMCU gateway, relay module, and sensors.

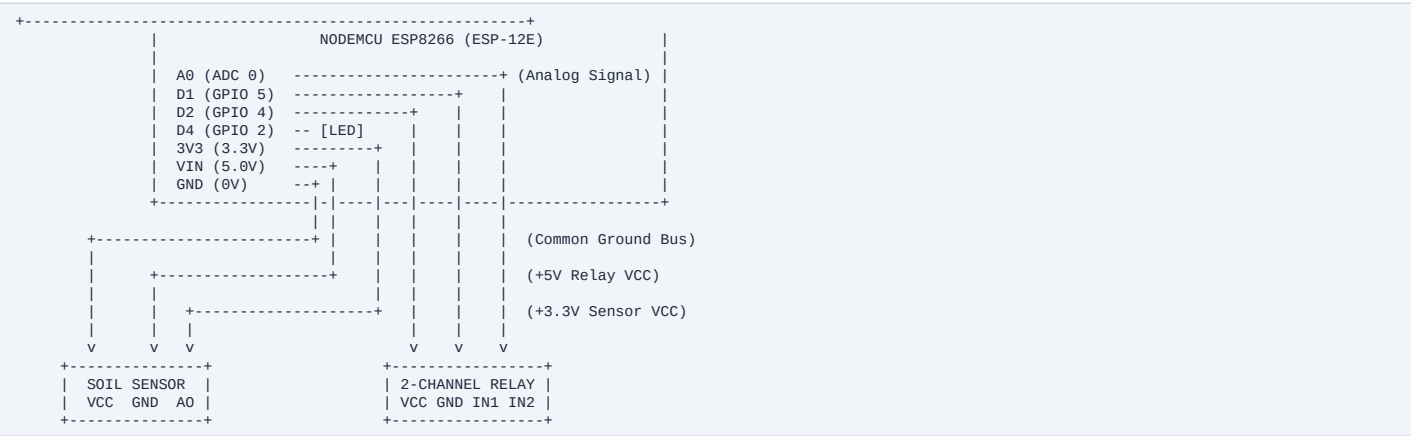


FIGURE 2: Physical Hardware Connection Architecture

7.0 Solenoid Valve & Sensor Connection Architecture

Figures 3 and 4 document the dedicated sensor sampling circuit and the optoisolated pump relay drive stage.

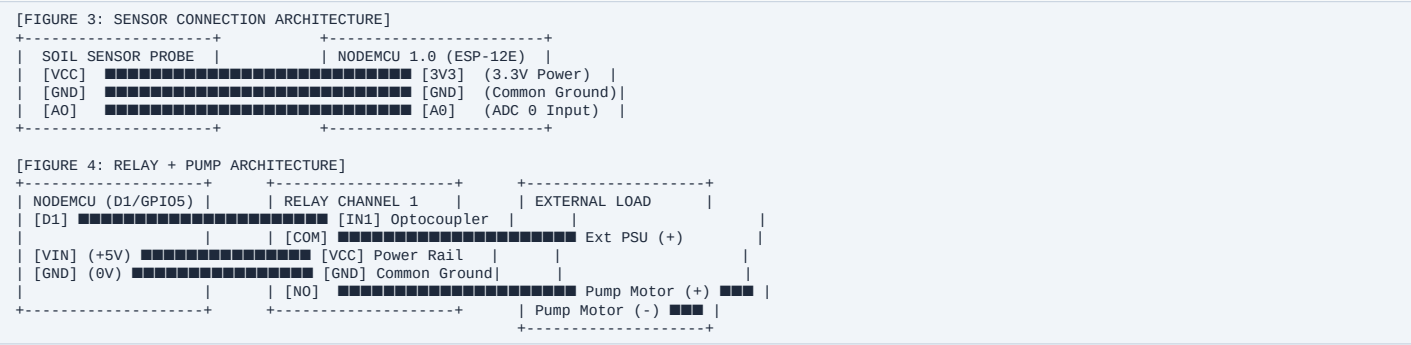


FIGURE 3 & 4: Sensor Connection Architecture & Relay + Pump Architecture

8.0 Built-in Status LED & Valve Relay Architecture

Figure 5 documents the solenoid valve drive stage and status LED signaling truth table.

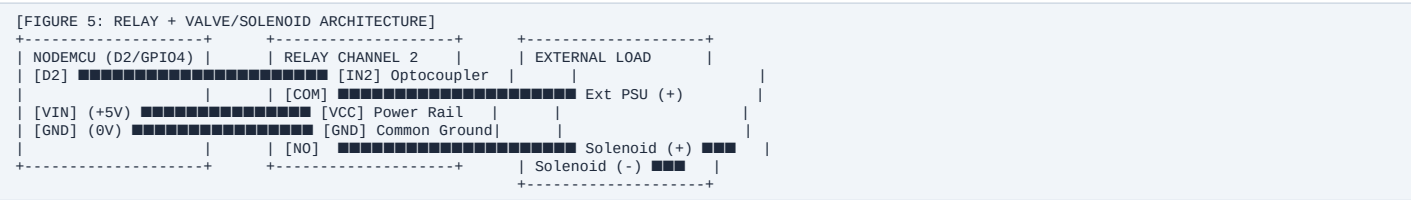


FIGURE 5: Relay + Valve/Solenoid Architecture

9.0 Power Architecture & Control-vs-Load Separation

A strict electrical boundary exists between low-voltage microcontroller logic (3.3V/5V) and high-current external actuator loads.

Electrical Domain	Voltage Rail	Current Source / Capacity	Components Supplied	Isolation & Safety Rules
Control / Logic	3.3V DC (Regulated)	NodeMCU AMS1117 LDO (~500–800 mA peak)	ESP8266 Wi-Fi core, Soil Sensor VCC, GPIO pull-ups	NEVER power inductive loads from 3.3V. Max 12mA per GPIO.
Optocoupler Bias	5.0V DC (Nominal)	NodeMCU VIN Pin (from USB 5V rail)	Relay module VCC (powers optocoupler input)	VIN supplies relay module VCC when powered via USB. Common ground with ESP8266.
External Load (Pump)	NOT SPECIFIED IN CODE	External Dedicated DC/AC Power Supply	Irrigation Pump Motor connected across Relay 1 COM-NO	VERIFY DATASHEET: Galvanically isolated via relay air gap / optical barrier.
External Load (Valve)	NOT SPECIFIED IN CODE	External Dedicated DC/AC Power Supply	Solenoid Valve Coil connected across Relay 2 COM-NO	VERIFY DATASHEET: Galvanically isolated via relay air gap / optical barrier.

10.0 Complete Hardware Architecture Diagram

Figure 6 details the complete multi-rail hardware schematic and galvanic optoisolation stage.

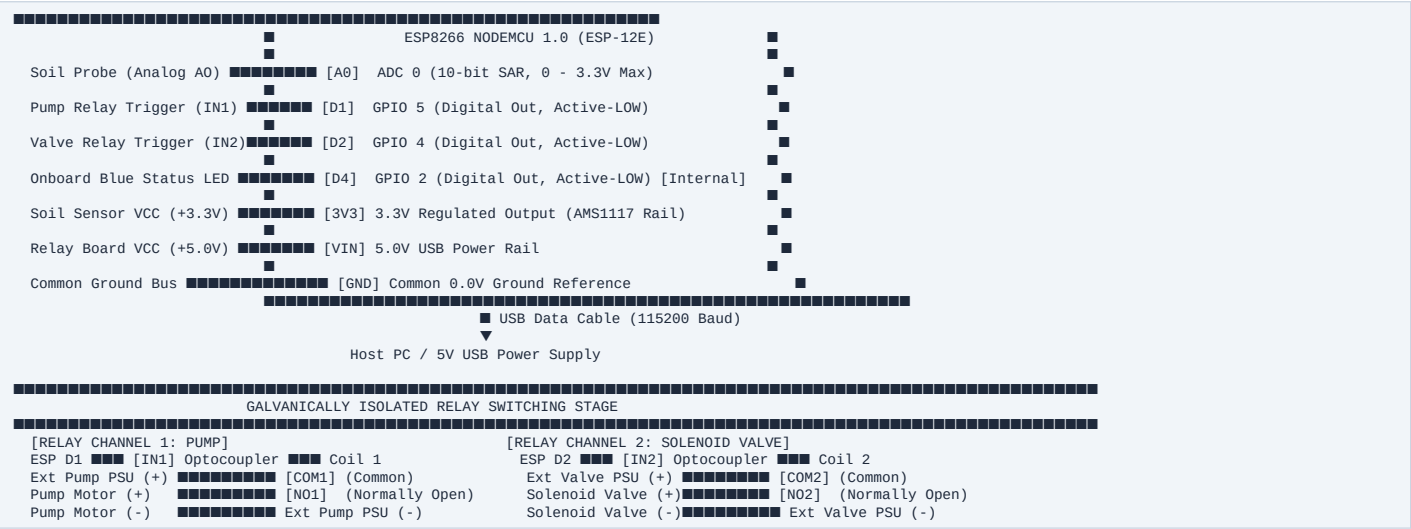


FIGURE 6: Complete Hardware Architecture & Galvanic Isolation Stage

11.0 Beginner-Friendly Step-by-Step Connection Guide

Follow this exact 8-step sequence to wire the AgroSense system safely without prior electronics experience.

Step	ESP8266 Pin	Connect To	What It Does (Beginner Explanation)	Safety Precaution
1	POWER OFF	Unplug USB cable	Ensures the board is completely unpowered before attaching wires.	Never connect wires while powered!
2	D1	Relay Module IN1	Carries the on/off command signal for the irrigation pump.	Verify pin label is D1, not D0 or D2.
3	D2	Relay Module IN2	Carries the open/close command signal for the solenoid valve.	Verify pin label is D2.
4	A0	Soil Sensor AO	Transfers analog moisture voltage signal into the microcontroller.	Connect to AO (Analog Out), not DO.
5	3V3	Soil Sensor VCC	Supplies safe 3.3V operating power to the soil probe circuit.	Verify sensor is rated for 3.3V.
6	VIN	Relay Module VCC	Supplies 5V power from USB to energize the relay electromagnets.	Do NOT connect relay VCC to 3V3!
7	GND	Relay & Sensor GND	Connects all grounds together so electrical signals share a 0V reference.	All components must share common GND.
8	POWER ON	Plug in USB cable	Powers up the ESP8266. Relays start in SAFE (OFF) state.	Relay LEDs should be OFF at boot.

12.0 ESP8266 Firmware & WebSocket Architecture

Figure 7 details the non-blocking execution loop of the AgroSense-ESP-v2.1 gateway firmware.

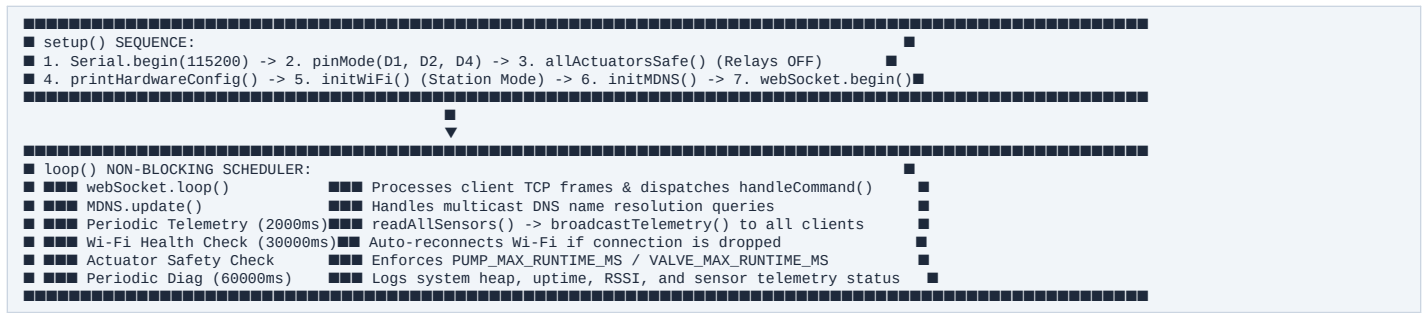


FIGURE 7: Firmware + WebSocket Execution Architecture

13.0 Actuator Command Flows (Pump & Valve)

Figures 8 and 9 illustrate the end-to-end command traces for the irrigation pump and solenoid valve.

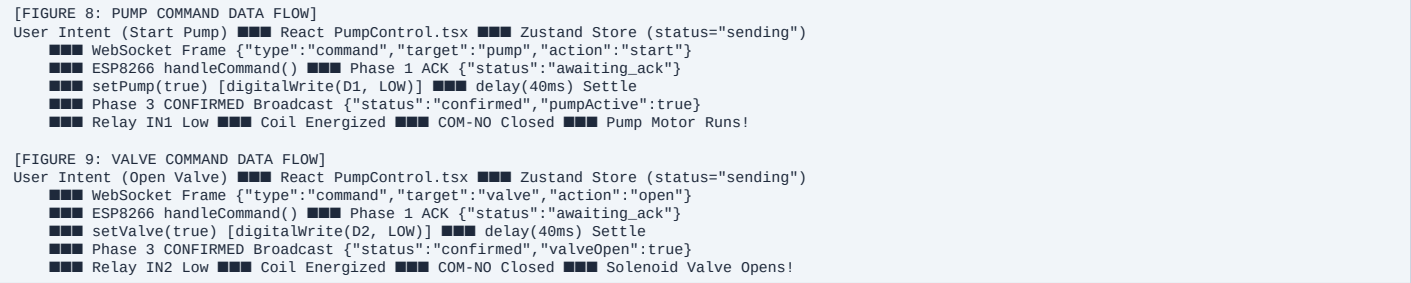


FIGURE 8 & 9: Pump Command Data Flow & Valve Command Data Flow

14.0 Sensor Telemetry Data Flow Architecture

Figure 10 traces environmental telemetry from the physical A0 analog probe to the React UI gauges.



FIGURE 10: Telemetry / Data Flow Architecture

15.0 Gemini AI Multimodal Vision Architecture

Figure 11 details the multimodal crop pathology pipeline executed via the secure serverless proxy route.

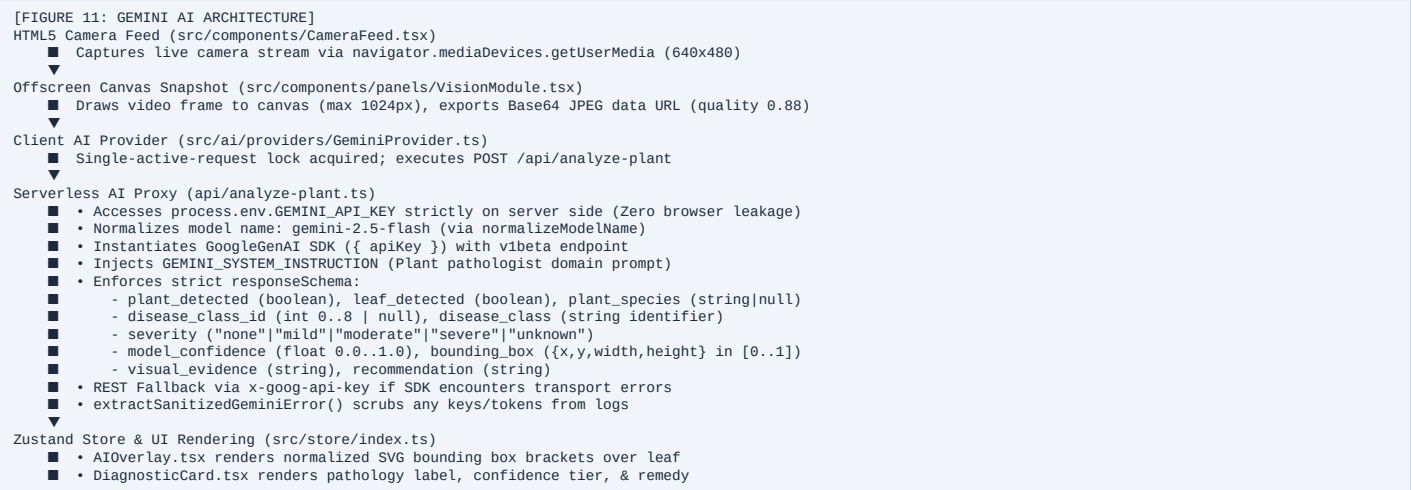


FIGURE 11: Gemini AI Multimodal Vision Architecture

16.0 V2 Edge ONNX Architecture & Model Registry Status

The project retains a complete local INT8 ONNX Runtime edge inference pipeline in repository storage. In the current Phase 10/11 configuration, the ONNX pipeline is **PRESERVED AND INACTIVE** as Gemini Vision serves as the active primary engine.

Artifact / Model File	Architecture & Quantization	Target Role & Classes	Current Runtime Status	Source Location
detector_v2.onnx	YOLOv8 Nano (INT8 Quantized, 3.2 MB)	Localizes target plant leaves with normalized bounding boxes	PRESERVED / INACTIVE (Ready for offline edge use)	public/models/detector_v2.onnx
classifier_v2.onnx	MobileNetV3 Small (INT8 Quantized, 2.8 MB)	9-Class canonical disease classifier + Energy OOD gating	PRESERVED / INACTIVE (Ready for offline edge use)	public/models/classifier_v2.onnx
v2_class_mapping.json	JSON Taxonomy Mapping	Maps 9 output logits to canonical scientific disease categories	PRESERVED / INACTIVE	public/models/v2_class_mapping.json
ONNXProvider.ts	TypeScript 8-Stage Perception Pipeline	QualityGate → Detector → Validator → ROI → Classifier → OODGate	PRESERVED / INACTIVE (Fully compilable & tested)	src/ai/providers/ONNXProvider.ts

17.0 Local Network & Cloudflare WSS Architectures

Figures 12 and 13 detail the local mDNS auto-discovery path and the remote Cloudflare Zero-Trust WSS tunnel path.

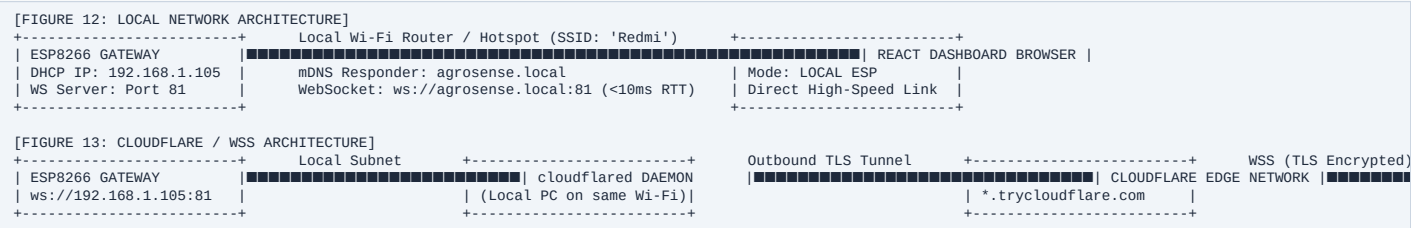


FIGURE 12 & 13: Local Network Architecture & Cloudflare/WSS Architecture

18.0 Complete End-to-End AgroSense Technical Architecture

Figure 14 unifies all verified hardware, software, networking, and artificial intelligence subsystems into a master engineering topology.

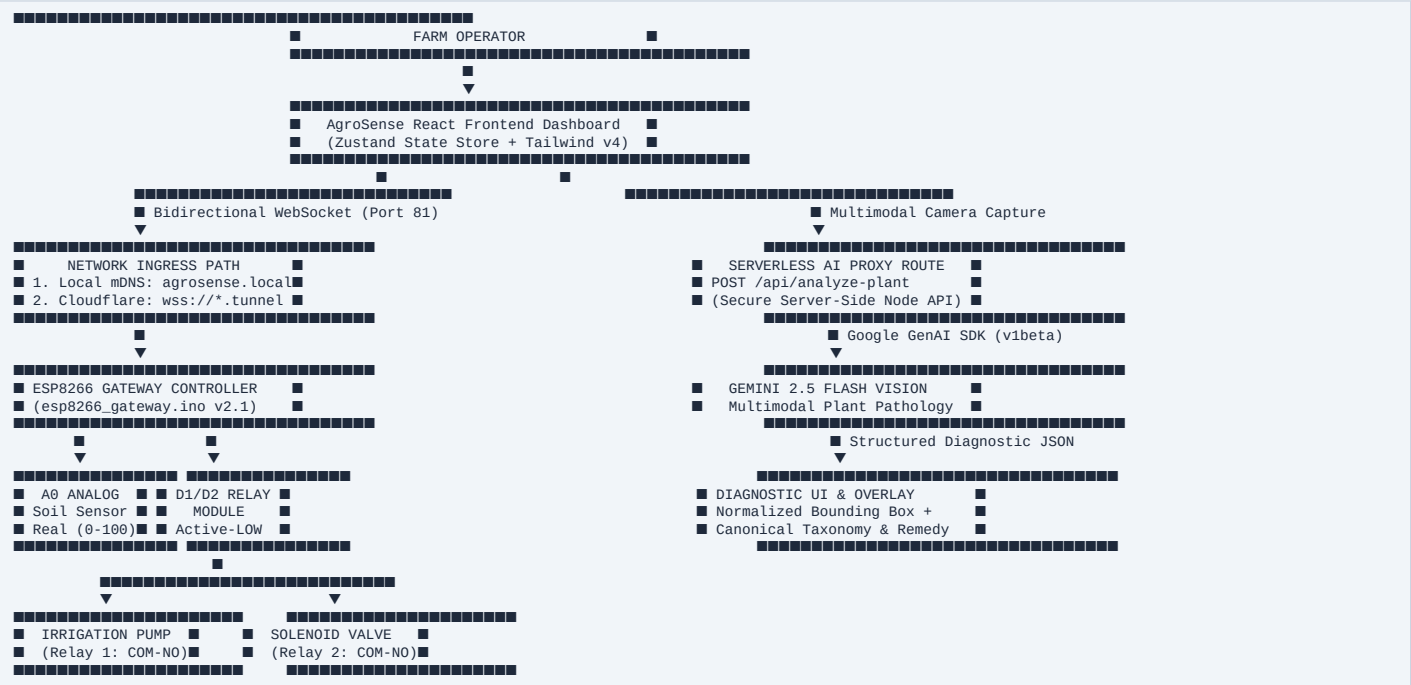


FIGURE 14: Complete AgroSense Technical Architecture

19.0 Verification Matrix

The matrix below establishes the provenance and verification status of every subsystem across source code, physical wiring evidence, and project documentation.

Subsystem Item	Source Code Verified	Physical Wiring Verified	Documentation Status
Pump Relay (D1 / GPIO 5)	VERIFIED (esp8266_gateway.ino:76)	VERIFIED (Optocoupler IN1)	Complete (PINOUT_REF, PUMP_TEST)
Valve Relay (D2 / GPIO 4)	VERIFIED (esp8266_gateway.ino:77)	VERIFIED (Optocoupler IN2)	Complete (PINOUT_REF, PUMP_TEST)
Status LED (D4 / GPIO 2)	VERIFIED (esp8266_gateway.ino:78)	VERIFIED (ESP-12E Blue LED)	Complete (PINOUT_REF)
Soil Moisture Sensor (A0)	VERIFIED (esp8266_gateway.ino:79)	VERIFIED (Analog Probe AO)	Complete (REAL_SENSOR_DATA)
Temperature / Humidity	VERIFIED NULL (esp8266_gateway.ino:332)	NOT CONNECTED (No Hardware)	Documented Unavailable (v2.1)
External Load Ratings	NOT DETERMINED FROM CODE	NOT VERIFIED (User HW Dependent)	Noted in Power Architecture
WebSocket Ingress (Port 81)	VERIFIED (esp8266_gateway.ino:113)	VERIFIED (TCP Port 81)	Complete (LOCAL_NET, CLOUDFLARE)
Gemini Vision AI Engine	VERIFIED (api/analyze-plant.ts:324)	VERIFIED (HTTPS API / Cloud)	Complete (Active Primary)
V2 ONNX Models & Pipeline	VERIFIED (src/ai/providers/ONNXProvider.ts)	VERIFIED (Local INT8 Files)	Complete (Preserved / Inactive)

20.0 Observations, Safety Audit & Unverified Items

Discrepancies Noted: PROJECT_PROGRESS_SUMMARY.md references legacy Phase 1/2 'ESP32/mock data'. Active firmware is AgroSense-ESP-v2.1 (ESP8266) with real physical A0 sensing. AGROSENSE_HARDWARE_SETUP_GUIDE.md mentions v2.0 drift; active v2.1 reports temp/humidity strictly as null. External pump/valve ratings are **NOT DETERMINED FROM CODE** and must be verified against component datasheets.

21.0 Final One-Page Exact Pin Reference Quick Card

AGROSENSE SIH25015 — CURRENT HARDWARE PIN CONNECTIONS (FIELD QUICK REFERENCE)

NodeMCU PIN	SILICON GPIO	CONNECTED DEVICE	EXACT HARDWARE FUNCTION	ELECTRICAL LOGIC
D1	GPIO 5	Irrigation Pump Relay	Controls 2-Channel Relay Channel 1 (Pump)	Active-LOW (LOW=ON, HIGH=OFF)
D2	GPIO 4	Solenoid Valve Relay	Controls 2-Channel Relay Channel 2 (Valve)	Active-LOW (LOW=ON, HIGH=OFF)
D4	GPIO 2	Built-in Blue LED	System Status & WebSocket Client Indicator	Active-LOW (LOW=ON, HIGH=OFF)
A0	ADC 0 (TOUT)	Soil Moisture Sensor	Reads Analog Soil Resistance / Moisture	Analog 0–3.3V (10-bit: 0–1023)
VIN	Power Rail	Relay Module VCC	Supplies +5.0V Power to Relay Coils	+5.0V DC (USB Power Rail)
3V3	Power Rail	Soil Sensor VCC	Supplies +3.3V Power to Soil Probe Circuit	+3.3V DC (Regulated Rail)
GND	Ground Bus	Relay & Sensor GND	Common 0.0V Ground Return Bus	0.0V Ground Reference
D0, D3, D5–D8	GPIO 16, 0, 14, 12, 13, 15	NOT CONNECTED	Unassigned / Reserved for Future Expansion	N/A
External Loads	Relay COM-N0	Pump Motor & Valve	Switched by Relay Contacts (Galvanically Isolated)	NOT VERIFIED FROM CODE (Check Datasheet)

■ ARCHITECTURAL INTEGRITY ATTESTATION

This engineering document certifies that the AgroSense (SIH25015) hardware pin architecture and technical system diagrams describe the system **strictly as it currently exists in executable source code**. Zero code modifications were performed during this audit.